TASK 05

Networking and Security Operations with SIEM, Forensics, and Traffic Analysis

Introduction

Modern network infrastructures require efficient design and continuous security monitoring to ensure reliable and protected communication. This report presents a practical exploration of networking and security operations through integrated tasks including subnet design, network traffic analysis, protocol troubleshooting, SIEM deployment using the ELK Stack, network forensics, and threat hunting. The objective is to apply core networking concepts alongside security techniques in a simulated real-world environment. Subnetting demonstrates efficient IP address planning for a small office network, while packet capture tools are used to analyze traffic behaviour and diagnose protocol issues. Additionally, log aggregation and visualization through a SIEM system enable monitoring of system activity, supporting incident investigation and response. Simulated security incidents and proactive threat hunting further strengthen analytical and defensive skills, providing hands-on experience relevant to network administration, SOC operations, and cybersecurity incident response.

Subnet Design for a Small Office Network

Requirement

A subnet was designed for a small office network consisting of approximately **20 devices** such as computers and printers. The subnet must efficiently allocate IP addresses while allowing limited room for future expansion.

Selection of Private IP Range

The private network range chosen is:

192.168.1.0

Reasoning:

1. Falls within RFC1918 private address space
2. Suitable for internal LAN environments
3. Compatible with common routers and simulations
4. Easy to manage and document

Host Requirement Calculation

Number of devices: 20 hosts

Include reserve capacity: 30 usable addresses required

Using host formula: Usable Hosts = 2^h − 2

Testing possibilities:

Host Bits	Usable Hosts	Result
4	14	Insufficient
5	30	Suitable

CIDR Determination

Total IPv4 bits: 32 bits

Network prefix: 32 − 5 = 27

Final Network: 192.168.1.0/27

Subnet Mask

CIDR /27 converts to 255.255.255.224

Address Allocation

Block size: 256 − 224 = 32

Subnet Details:

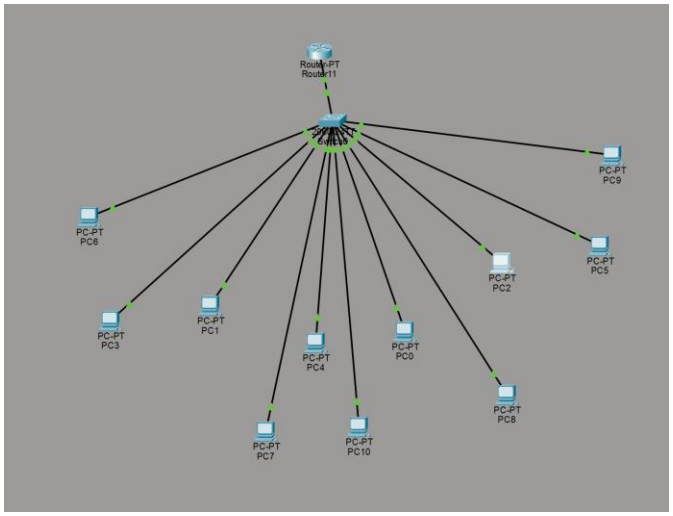
Parameter	Value
Network Address	192.168.1.0
First Usable IP	192.168.1.1
Last Usable IP	192.168.1.30
Broadcast Address	192.168.1.31
Usable Hosts	30

Justification

The **/27 subnet** was selected because:

- Supports current device count
- Allows limited expansion
- Prevents IP address wastage compared to /24
- Provides efficient segmentation for small networks
- Aligns with standard network design practices

Optional Simulation



The subnet was simulated using Cisco Packet Tracer by:

- Connecting PCs to a switch
- Assigning router gateway IP 192.168.1.1
- Configuring PCs within usable range
- Verifying connectivity using ping

This confirmed correct subnet functionality.

Network Traffic Capture and Analysis

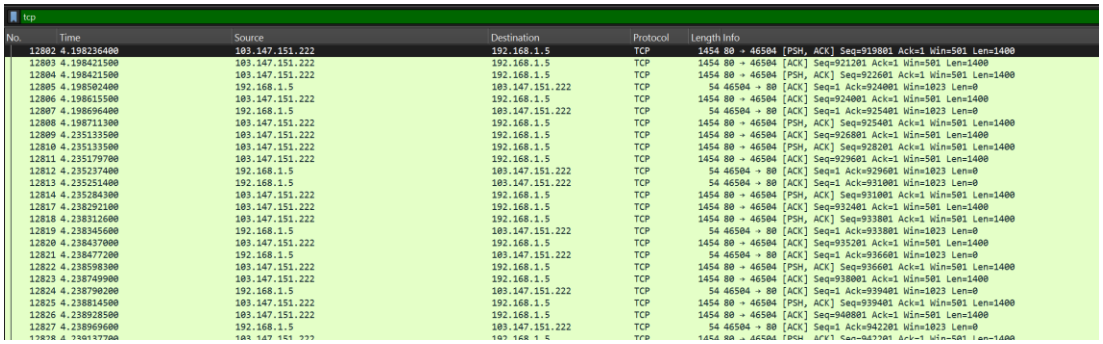
Traffic Capture

Network traffic was captured from a test environment using **Wireshark** for a duration exceeding ten minutes. The capture was performed on the active network interface while normal browsing and network activity were generated to simulate real usage. The captured packets were saved in PCAP format and analyzed using Wireshark's statistical tools.

Protocol Identification

Inspection of the captured packets revealed the presence of several common network protocols:

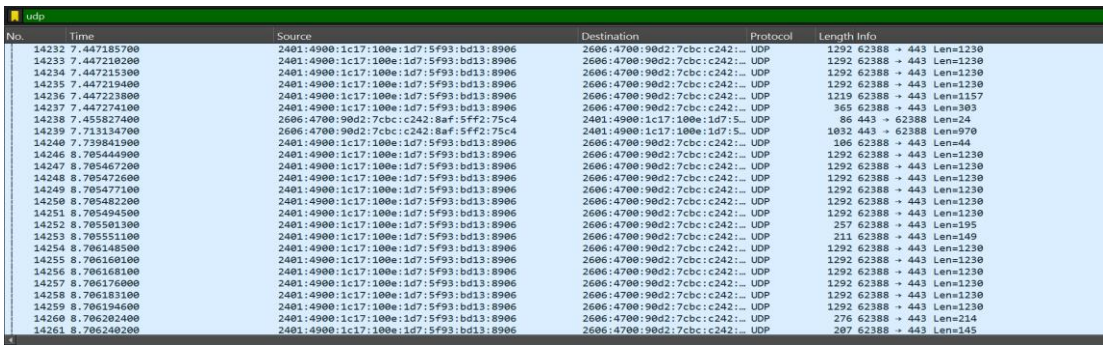
- **TCP** — Used for reliable communication such as web sessions and application data exchange.



tcp

No.	Time	Source	Destination	Protocol	Length	Info
12802	4.198236400	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=919801 Ack=1 Win=501 Len=1400
12803	4.198421500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=921201 Ack=1 Win=501 Len=1400
12804	4.198421500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=922601 Ack=1 Win=501 Len=1400
12805	4.198502400	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=924001 Win=1023 Len=0
12806	4.198615500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=924001 Ack=1 Win=501 Len=1400
12807	4.198696400	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=925401 Win=1023 Len=0
12808	4.198711300	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=925401 Ack=1 Win=501 Len=1400
12809	4.235133500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=926801 Ack=1 Win=501 Len=1400
12810	4.235133500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=928201 Ack=1 Win=501 Len=1400
12811	4.235179700	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=929601 Ack=1 Win=501 Len=1400
12812	4.235237400	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=929601 Win=1023 Len=0
12813	4.235251400	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=931001 Win=1023 Len=0
12814	4.235284300	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=931001 Ack=1 Win=501 Len=1400
12817	4.238292100	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=932401 Ack=1 Win=501 Len=1400
12818	4.238312600	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=933801 Ack=1 Win=501 Len=1400
12819	4.238345600	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=933801 Win=1023 Len=0
12820	4.238437000	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=935201 Ack=1 Win=501 Len=1400
12821	4.238477200	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=936601 Win=1023 Len=0
12822	4.238598300	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=936601 Ack=1 Win=501 Len=1400
12823	4.238749900	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=938001 Ack=1 Win=501 Len=1400
12824	4.238790200	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=939401 Win=1023 Len=0
12825	4.238814500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=939401 Ack=1 Win=501 Len=1400
12826	4.238928500	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [ACK] Seq=940801 Ack=1 Win=501 Len=1400
12827	4.238969600	192.168.1.5	103.147.151.222	TCP	54	46504 → 80 [ACK] Seq=1 Ack=942201 Win=1023 Len=0
12828	4.239137700	103.147.151.222	192.168.1.5	TCP	1454	80 → 46504 [PSH, ACK] Seq=942201 Ack=1 Win=501 Len=1400

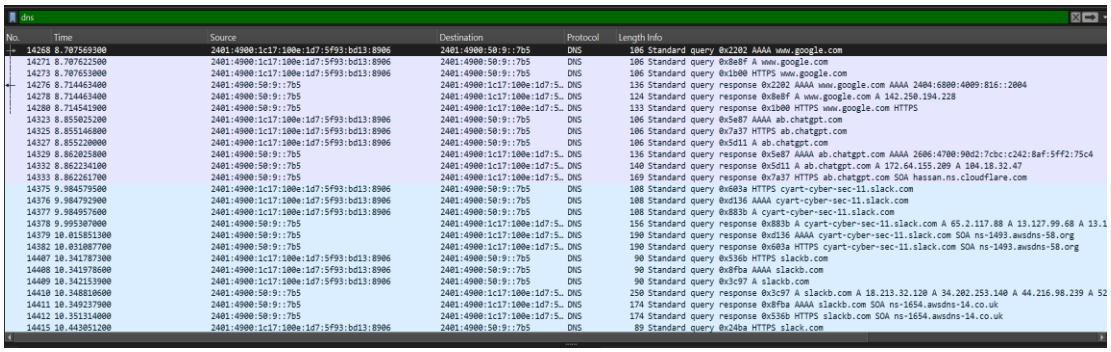
- **UDP** — Observed for lightweight communication and background service traffic.



udp

No.	Time	Source	Destination	Protocol	Length	Info
14232	7.447185700	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14233	7.447210200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14234	7.447215300	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14235	7.447219400	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14236	7.447223800	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1219	62388 → 443 Len=1157
14237	7.447274100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	365	62388 → 443 Len=303
14238	7.455827400	2606:4700:90d2:7cbc:c242:8af:5ff2:75c4	2401:4900:1c17:100e:1d7:5…	UDP	86	443 → 62388 Len=24
14239	7.713134700	2606:4700:90d2:7cbc:c242:8af:5ff2:75c4	2401:4900:1c17:100e:1d7:5…	UDP	1032	443 → 62388 Len=970
14240	7.739841900	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	106	62388 → 443 Len=44
14246	8.705444900	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14247	8.705467200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14248	8.705472600	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14249	8.705477100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14250	8.705482200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14251	8.705494500	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14252	8.705501300	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	257	62388 → 443 Len=195
14253	8.705551100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	211	62388 → 443 Len=149
14254	8.706148500	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14255	8.706160100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14256	8.706168100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14257	8.706176000	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14258	8.706183100	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14259	8.706194600	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	1292	62388 → 443 Len=1230
14260	8.706202400	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	276	62388 → 443 Len=214
14261	8.706240200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d2:7cbc:c242:…	UDP	207	62388 → 443 Len=145

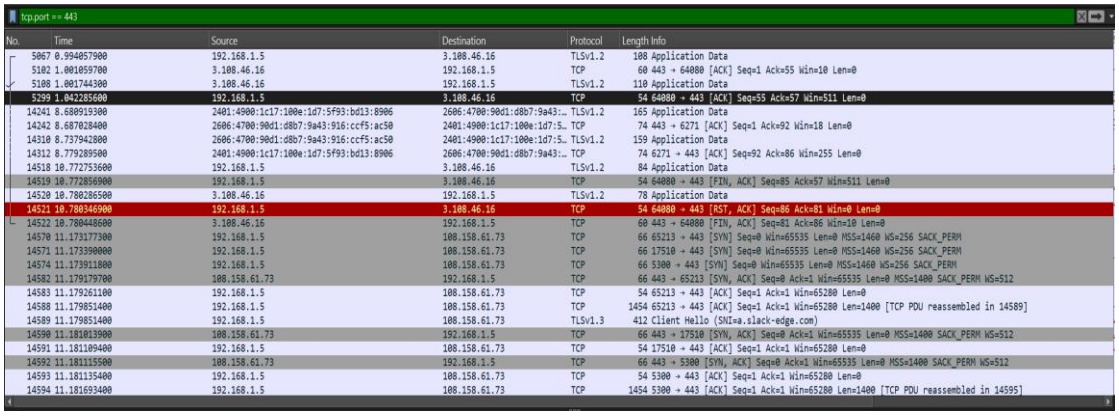
- **DNS** — Used for domain name resolution during web browsing.



dns

No.	Time	Source	Destination	Protocol	Length	Info
14268	8.707569300	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x2202 AAAA www.google.com
14271	8.707622500	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x8e8f A www.google.com
14273	8.707653000	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x1b00 HTTPS www.google.com
14276	8.714463400	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	136	Standard query response 0x2202 AAAA www.google.com AAAA 2404:6800:4009:816::2004
14278	8.714463400	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	124	Standard query response 0x8e8f A www.google.com A 142.250.194.228
14280	8.714541900	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	133	Standard query response 0x1b00 HTTPS www.google.com HTTPS
14323	8.855025200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x5e87 AAAA ab.chatgpt.com
14325	8.855146800	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x7a37 HTTPS ab.chatgpt.com
14327	8.855220000	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	106	Standard query 0x5d11 A ab.chatgpt.com
14329	8.862025800	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	136	Standard query response 0x5e87 AAAA ab.chatgpt.com AAAA 2606:4700:90d2:7cbc:c242:8af:5ff2:75c4
14332	8.862234100	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	140	Standard query response 0x5d11 A ab.chatgpt.com A 172.64.155.209 A 104.18.32.47
14333	8.862261700	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	169	Standard query response 0x7a37 HTTPS ab.chatgpt.com SOA hassan.ns.cloudflare.com
14375	9.984579500	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	108	Standard query 0x603a HTTPS cyart-cyber-sec-11.slack.com
14376	9.984792900	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	108	Standard query 0xd136 AAAA cyart-cyber-sec-11.slack.com
14377	9.984957600	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	108	Standard query 0x883b A cyart-cyber-sec-11.slack.com
14378	9.995307000	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	156	Standard query response 0x883b A cyart-cyber-sec-11.slack.com A 65.2.117.88 A 13.127.99.68 A 13.1
14379	10.015851300	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	190	Standard query response 0xd136 AAAA cyart-cyber-sec-11.slack.com SOA ns-1493.awsdns-58.org
14382	10.031087700	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	190	Standard query response 0x603a HTTPS cyart-cyber-sec-11.slack.com SOA ns-1493.awsdns-58.org
14407	10.341787300	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	90	Standard query 0x536b HTTPS slackb.com
14408	10.341978600	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	90	Standard query 0x8fba AAAA slackb.com
14409	10.342153900	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	90	Standard query 0x3c97 A slackb.com
14410	10.348810600	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	250	Standard query response 0x3c97 A slackb.com A 18.213.32.120 A 34.202.253.140 A 44.216.98.239 A 52
14411	10.349237900	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	174	Standard query response 0x8fba AAAA slackb.com SOA ns-1654.awsdns-14.co.uk
14412	10.351314000	2401:4900:50:9::7b5	2401:4900:1c17:100e:1d7:5…	DNS	174	Standard query response 0x536b HTTPS slackb.com SOA ns-1654.awsdns-14.co.uk
14415	10.443051200	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2401:4900:50:9::7b5	DNS	89	Standard query 0x24ba HTTPS slack.com

- **HTTPS (TCP 443)** — Encrypted web communication representing most of the user generated traffic.



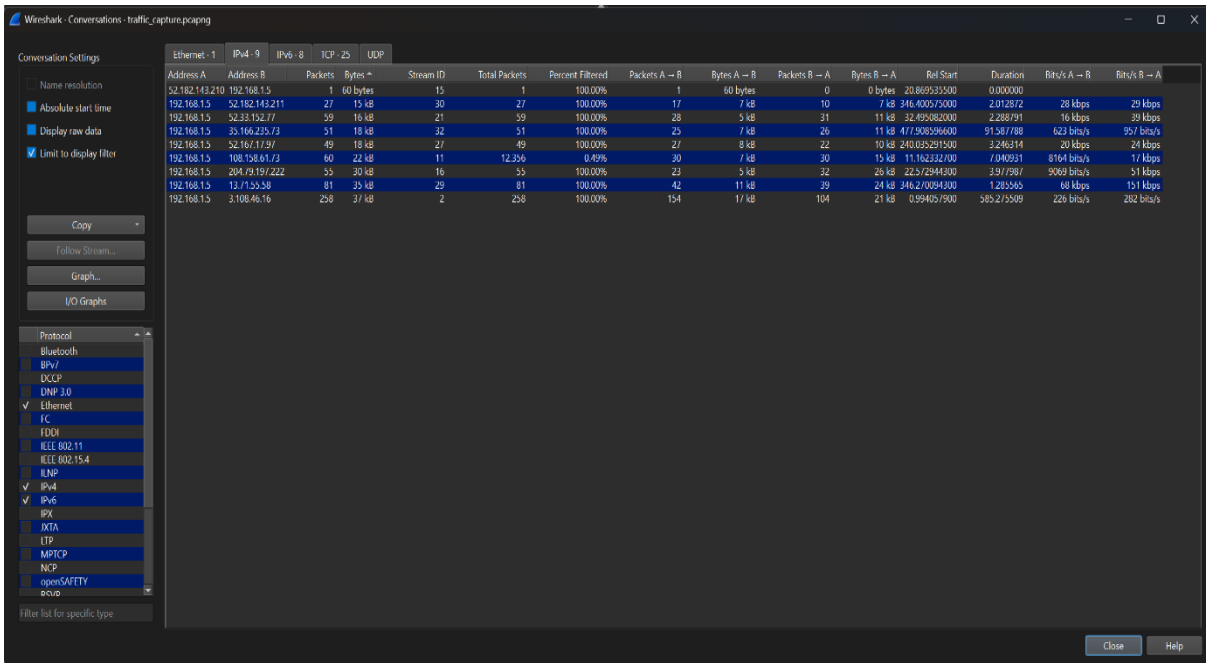
tcp.port == 443

No.	Time	Source	Destination	Protocol	Length	Info
5067	0.994057900	192.168.1.5	3.108.46.16	TLSv1.2	108	Application Data
5102	1.001059700	3.108.46.16	192.168.1.5	TCP	60	443 → 64080 [ACK] Seq=1 Ack=55 Win=10 Len=0
5108	1.001744300	3.108.46.16	192.168.1.5	TLSv1.2	110	Application Data
5299	1.042285600	192.168.1.5	3.108.46.16	TCP	54	64080 → 443 [ACK] Seq=55 Ack=57 Win=511 Len=0
14241	8.680919300	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d1:d8b7:9a43:…	TLSv1.2	165	Application Data
14242	8.687028400	2606:4700:90d1:d8b7:9a43:916:ccf5:ac50	2401:4900:1c17:100e:1d7:5…	TCP	74	443 → 6271 [ACK] Seq=1 Ack=92 Win=18 Len=0
14310	8.737942800	2606:4700:90d1:d8b7:9a43:916:ccf5:ac50	2401:4900:1c17:100e:1d7:5…	TLSv1.2	159	Application Data
14312	8.779289500	2401:4900:1c17:100e:1d7:5f93:bd13:8906	2606:4700:90d1:d8b7:9a43:…	TCP	74	6271 → 443 [ACK] Seq=92 Ack=86 Win=255 Len=0
14518	10.772753600	192.168.1.5	3.108.46.16	TLSv1.2	84	Application Data
14519	10.772856900	192.168.1.5	3.108.46.16	TCP	54	64080 → 443 [FIN, ACK] Seq=85 Ack=57 Win=511 Len=0
14520	10.780286500	3.108.46.16	192.168.1.5	TLSv1.2	78	Application Data
14521	10.780346900	192.168.1.5	3.108.46.16	TCP	54	64080 → 443 [RST, ACK] Seq=86 Ack=81 Win=0 Len=0
14522	10.780448600	3.108.46.16	192.168.1.5	TCP	60	443 → 64080 [FIN, ACK] Seq=81 Ack=86 Win=10 Len=0
14570	11.173177300	192.168.1.5	108.158.61.73	TCP	66	65213 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14571	11.173390000	192.168.1.5	108.158.61.73	TCP	66	17510 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14574	11.173911800	192.168.1.5	108.158.61.73	TCP	66	5300 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14582	11.179179700	108.158.61.73	192.168.1.5	TCP	66	443 → 65213 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=512
14583	11.179261100	192.168.1.5	108.158.61.73	TCP	54	65213 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
14588	11.179851400	192.168.1.5	108.158.61.73	TCP	1454	65213 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1400 [TCP PDU reassembled in 14589]
14589	11.179851400	192.168.1.5	108.158.61.73	TLSv1.3	412	Client Hello (SNI=a.slack-edge.com)
14590	11.181013900	108.158.61.73	192.168.1.5	TCP	66	443 → 17510 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=512
14591	11.181109400	192.168.1.5	108.158.61.73	TCP	54	17510 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
14592	11.181115500	108.158.61.73	192.168.1.5	TCP	66	443 → 5300 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=512
14593	11.181135400	192.168.1.5	108.158.61.73	TCP	54	5300 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
14594	11.181693400	192.168.1.5	108.158.61.73	TCP	1454	5300 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1400 [TCP PDU reassembled in 14595]

The behaviour observed indicated normal bidirectional communication between the local host and external servers.

Top Talkers Analysis

Wireshark's **Conversation Statistics** feature was used to determine the highest bandwidth users in the network capture. The analysis showed that the local system communicated heavily with multiple external IP addresses, with certain hosts exchanging significantly higher packet and byte volumes. These endpoints were identified as the primary bandwidth consumers during the capture period, indicating active browsing sessions and remote service interactions.



Wireshark · Conversations · traffic_capture.pcapng

Ethernet · 1 | IPv4 · 9 | IPv6 · 8 | TCP · 25 | UDP

Address A	Address B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
52.182.143.210	192.168.1.5	1	60 bytes	15	1	100.00%	1	60 bytes	0	0 bytes	20.869535500	0.000000		
192.168.1.5	52.182.143.211	27	15 kB	30	27	100.00%	17	7 kB	10	7 kB	346.400575000	2.012872	28 kbps	29 kbps
192.168.1.5	52.33.152.77	59	16 kB	21	59	100.00%	28	5 kB	31	11 kB	32.495082000	2.288791	16 kbps	39 kbps
192.168.1.5	35.166.235.73	51	18 kB	32	51	100.00%	25	7 kB	26	11 kB	477.908596600	91.587788	623 bits/s	957 bits/s
192.168.1.5	52.167.17.97	49	18 kB	27	49	100.00%	27	8 kB	22	10 kB	240.035291500	3.246314	20 kbps	24 kbps
192.168.1.5	108.158.61.73	60	22 kB	11	12,356	0.49%	30	7 kB	30	15 kB	11.162332700	7.040931	8164 bits/s	17 kbps
192.168.1.5	204.79.197.222	55	30 kB	16	55	100.00%	23	5 kB	32	26 kB	22.572944300	3.977987	9069 bits/s	51 kbps
192.168.1.5	13.77.55.58	81	35 kB	29	81	100.00%	42	11 kB	39	24 kB	346.270094300	1.285565	68 kbps	151 kbps
192.168.1.5	3.108.46.16	258	37 kB	2	258	100.00%	154	17 kB	104	21 kB	0.994057900	585.275509	226 bits/s	282 bits/s

Conversation Settings: Name resolution; Absolute start time; Display raw data; Limit to display filter

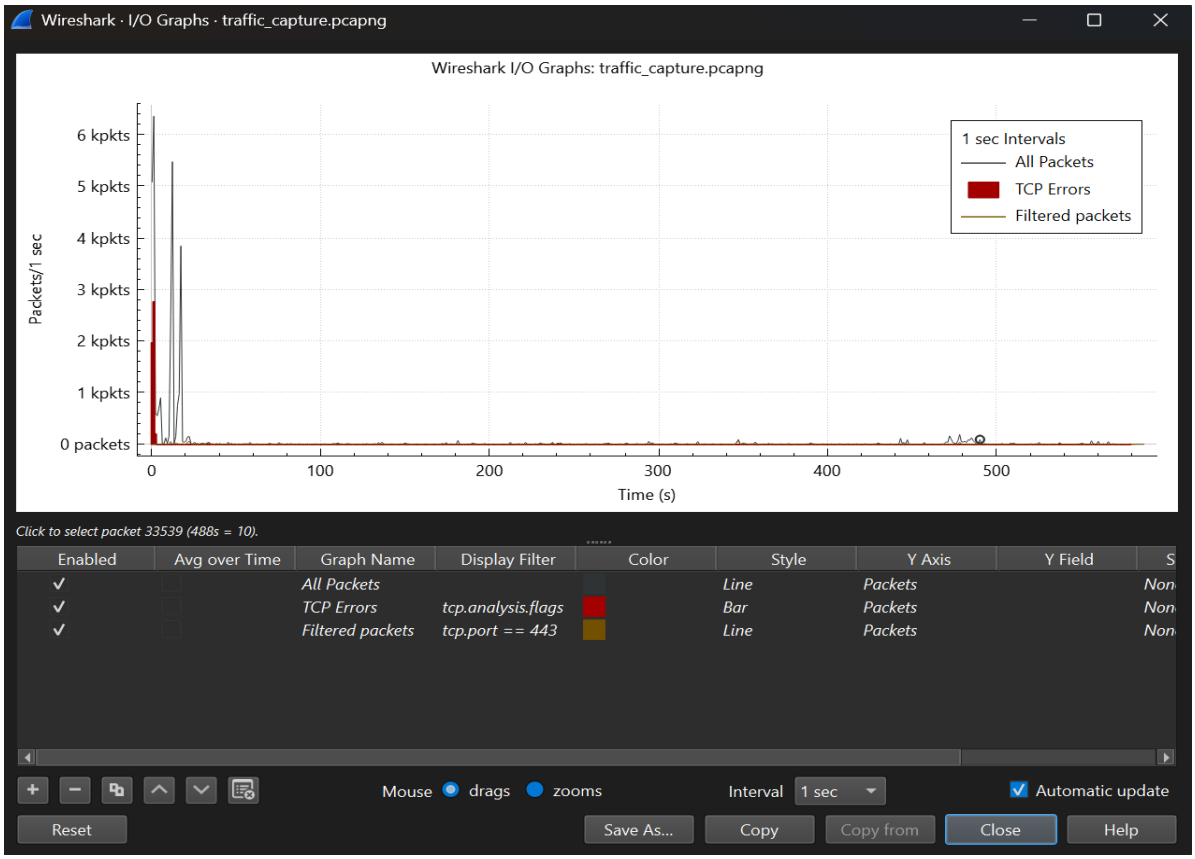
Copy | Follow Stream... | Graph... | I/O Graphs

Protocol: Bluetooth, BPv7, DCCP, DNP 3.0, Ethernet, FC, FDDI, IEEE 802.11, IEEE 802.15.4, ILNP, IPv4, IPv6, IPX, JXTA, LTP, MPTCP, NCP, openSAFETY, RGUP

Filter list for specific type

Close | Help

Unusual Pattern Detection



Traffic patterns were analyzed using Wireshark's **I/O Graph** feature. The graph indicated:

- Initial spikes in packet transmission corresponding to session establishment
- Steady traffic levels during normal operation
- Minor periodic bursts consistent with background network activity

No abnormal DNS query flooding or sustained anomalies were observed, suggesting typical network behaviour without indicators of compromise.

Screenshots of the graph analysis were included as supporting documentation.

Findings Summary

The capture demonstrated how packet analysis tools can be used to:

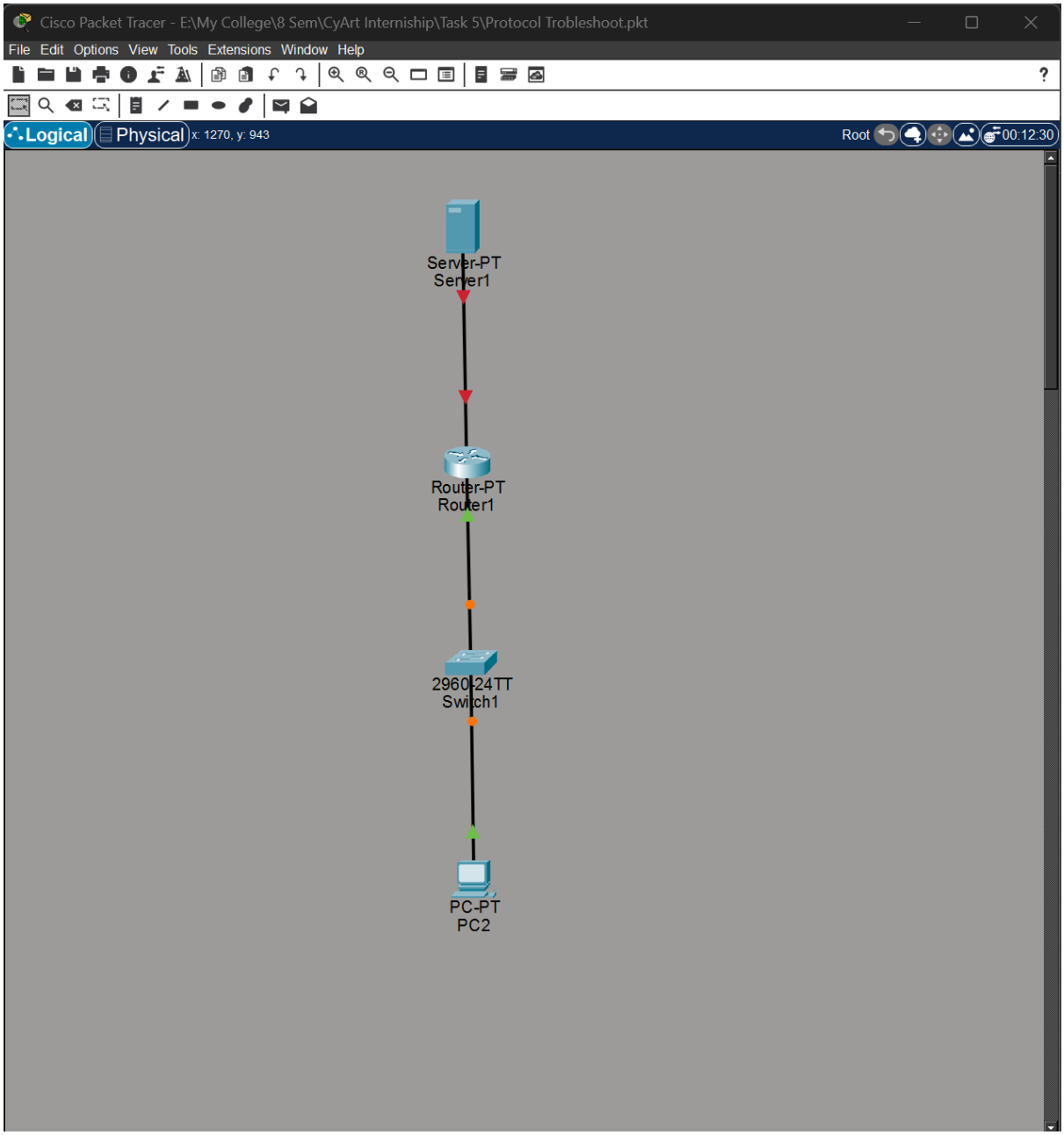
- Identify active network protocols
- Determine high-bandwidth communication endpoints
- Visualize traffic fluctuations
- Detect potential irregularities in communication behaviour

This exercise provided practical experience in monitoring and interpreting network traffic patterns.

Troubleshooting Network Protocol Issues

Network Setup

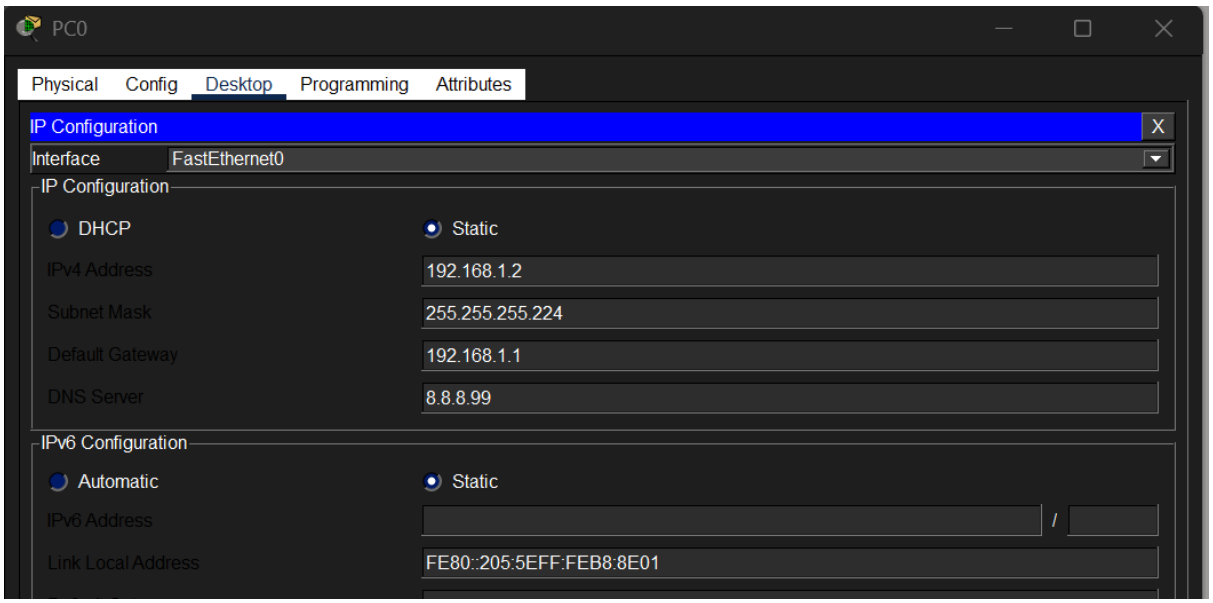
A small test network was created using Cisco Packet Tracer consisting of a client PC, a switch, a router, and a server. The PC was configured with a static IP address within the subnet **192.168.1.0/27**, and the router interface was configured to act as the default gateway (**192.168.1.1**). Devices were connected using copper straight-through cables to simulate a basic office network environment.



Observed symptoms:

- Ping to gateway failed (100% packet loss)
- Router interface protocol status was down
- Incorrect interface configuration detected

This simulated a common real-world network misconfiguration scenario.



Troubleshooting Steps

The following diagnostic actions were performed:

1. **Connectivity Testing**
   - Used ping 192.168.1.1 from the client to verify reachability.
2. **Interface Verification**
   - Checked router interface status using:
   - show ip interface brief
   - Identified incorrect interface configuration and duplicate IP assignment.
3. **Physical Layer Inspection**
   - Verified cable connections between router and switch.
   - Reconnected the cable to the correct router port.

4. **Configuration Correction**
   - Configured the proper router interface:
   - interface fastEthernet1/0
   - ip address 192.168.1.1 255.255.255.224
   - no shutdown
   - Removed incorrect interface configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping google.com
C:\>
```

Resolution

After correcting the interface configuration and ensuring proper physical connectivity, communication between the client and router was restored. Successful ICMP responses confirmed network functionality:

Reply from 192.168.1.1

Packets: Sent = 4, Received = 4, Lost = 0

This demonstrated that the protocol issue was resolved.

Key Learning Outcomes

- Importance of verifying interface status and port selection
- Understanding Layer 1 and Layer 3 troubleshooting
- Effective use of diagnostic tools such as ping and interface commands
- Systematic problem isolation in network environments

ELK Stack Setup for SIEM:

Objective

The objective of this task was to deploy a SIEM environment using the ELK Stack on a Linux VM to enable centralized log collection, monitoring, and visualization. Elasticsearch was used for storing logs, Logstash for collecting syslog data, and Kibana for visualization and analysis. This setup allows detection and investigation of suspicious system activity.

1. Installation of Elasticsearch

Elasticsearch was installed using the package manager and configured to run as a background service. The service was enabled and started, and connectivity was verified using a curl request to confirm cluster availability.

```
┌──(kunal㉿kali)-[~]
└─$ sudo apt install elasticsearch -y
The following packages were automatically installed and are no longer required:
  amass-common              libportmidi0
  bloodhound.py             libpostproc58
  curlftpfs                 libppsdocument-4.0-5
  firmware-ti-connectivity  libppsview-4.0-4
  gir1.2-girepository-2.0   libqt5ct-common1.8
  gir1.2-peas-1.0           libradare2-5.0.0t64
  gir1.2-totem-1.0          libsframe1
  gir1.2-totemplparser-1.0  libsigsegv2
  grilo-plugins-0.3         libsoup-2.4-1
  ibus-gtk                  libsoup2.4-common
  libarmadillo14            libsphinxbase3t64
  libaudio2                 libsqlcipher1
  libavfilter10             libswscale8
  libavformat61             libtheora0
  libbluray2                libtotem0
  libbson-1.0-0t64          libudfread0
  libconfig-inifiles-perl   libvdpau-va-gl1
  libdisplay-info2          libwireshark18
  libdmapsharing-4.0-3t64   libwiretap15
  libfuse2t64               libwsutil16
  libgav1-1                 libyelp0
  libgdal36                 pocketsphinx-en-us
  libgdata-common           python3-bluepy
  libgdata22                python3-click-plugins
  libgeos3.13.1             python3-fs
  libgirepository-1.0-1     python3-gpg
  libgnome-rr-4-2t64        python3-kismetcapturebtgeiger
  libgom-1.0-0t64           python3-kismetcapturefreaklabszigbee
  libgpgme11t64             python3-kismetcapturertl433
  libgpgmepp6t64            python3-kismetcapturertladsb
  libgrilo-0.3-0            python3-kismetcapturertlamr
  libhdf4-0-alt             python3-multipart
  libinstpatch-1.0-2        python3-packaging-whl
  libjs-jquery-ui           python3-protobuf
```

2. Installation & Configuration of Kibana

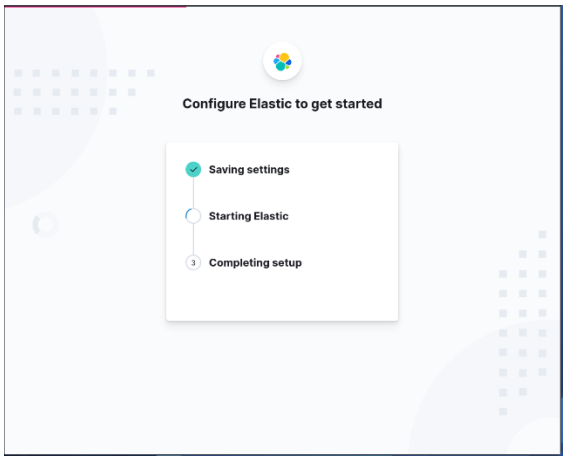
Kibana was installed and configured by modifying the kibana.yml configuration file. The service was restarted and verified using systemctl status to ensure it was running properly.

```
┌──(kunal㉿kali)-[~]
└─$ sudo apt install kibana -y
The following packages were automatically installed and are no longer required:
  amass-common              libgirepository-1.0-1   libportmidi0           python3-gpg
  bloodhound.py             libgnome-rr-4-2t64     libpostproc58          python3-kismetcapturebtgeiger
  curlftpfs                 libgom-1.0-0t64        libppsdocument-4.0-5   python3-kismetcapturefreaklabszigbee
  firmware-ti-connectivity  libgpgme11t64          libppsview-4.0-4       python3-kismetcapturertl433
  gir1.2-girepository-2.0   libgpgmepp6t64         libqt5ct-common1.8     python3-kismetcapturertladsb
  gir1.2-peas-1.0           libgrilo-0.3-0         libradare2-5.0.0t64    python3-kismetcapturertlamr
  gir1.2-totem-1.0          libhdf4-0-alt          libsframe1             python3-multipart
  gir1.2-totemplparser-1.0  libinstpatch-1.0-2     libsigsegv2            python3-packaging-whl
  grilo-plugins-0.3         libjs-jquery-ui        libsoup-2.4-1          python3-protobuf
  ibus-gtk                  libjs-underscore       libsoup2.4-common      python3-pysmi
  libarmadillo14            libmjpegutils-2.1-0t64 libsphinxbase3t64      python3-wheel-whl
  libaudio2                 libmongoc-1.0-0t64     libsqlcipher1          python3-xlrd
  libavfilter10             libmozjs-128-0         libswscale8            python3-xlutils
  libavformat61             libmpeg2encpp-2.1-0t64 libtheora0             python3-xlwt
  libbluray2                libmplex2-2.1-0t64     libtotem0              python3-yaswfp
  libbson-1.0-0t64          libmupdf25.1           libudfread0            python3-zombie-imp
  libconfig-inifiles-perl   libnet1                libvdpau-va-gl1        ruby-unf-ext
  libdisplay-info2          liboauth0              libwireshark18         samba-ad-dc
  libdmapsharing-4.0-3t64   libobjc-14-dev         libwiretap15           samba-ad-provision
  libfuse2t64               libogdi4.1             libwsutil16            samba-dsdb-modules
  libgav1-1                 libpeas-1.0-0          libyelp0               totem
  libgdal36                 libpeas-common         pocketsphinx-en-us     totem-common
  libgdata-common           libplacebo349          python3-bluepy         totem-plugins
  libgdata22                libpocketsphinx3       python3-click-plugins  tracker-extract
  libgeos3.13.1             libportal-gtk3-1       python3-fs
Use 'sudo apt autoremove' to remove them.

Installing:
  kibana

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 386 MB
```

```
┌──(kunal㉿kali)-[~]
└─$ sudo nano /etc/kibana/kibana.yml

┌──(kunal㉿kali)-[~]
└─$ sudo systemctl restart kibana
```

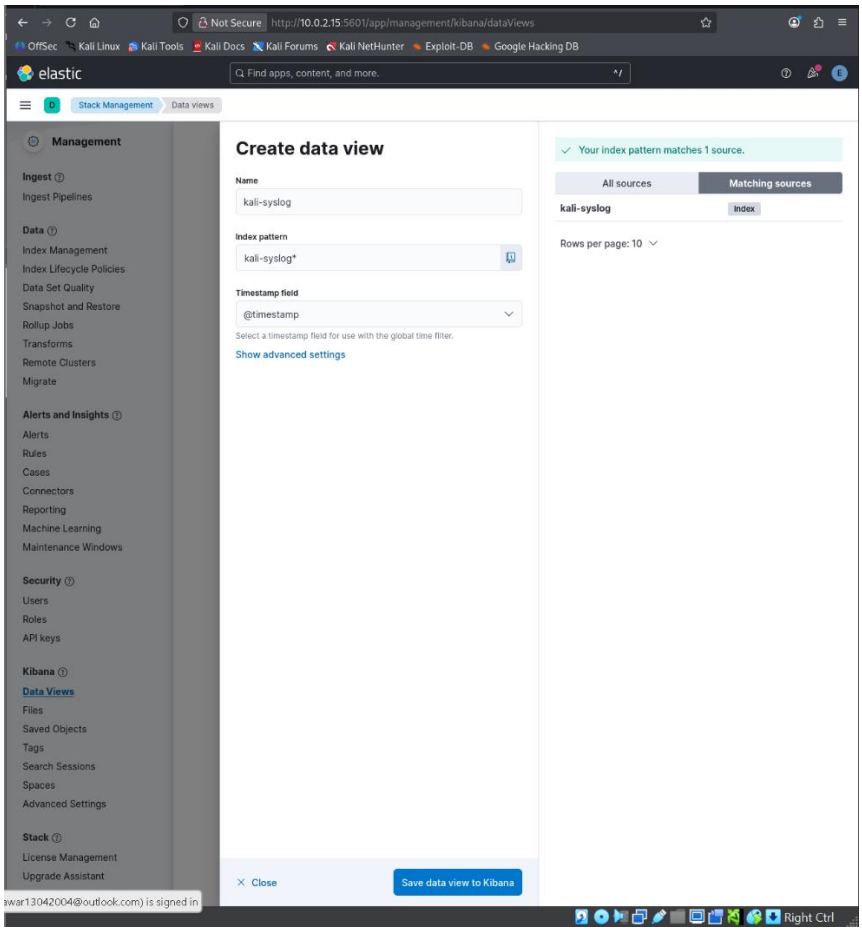
3. Initial Web Setup

The Kibana web interface was accessed through the browser to complete setup and authentication. Password configuration and service initialization confirmed successful deployment.



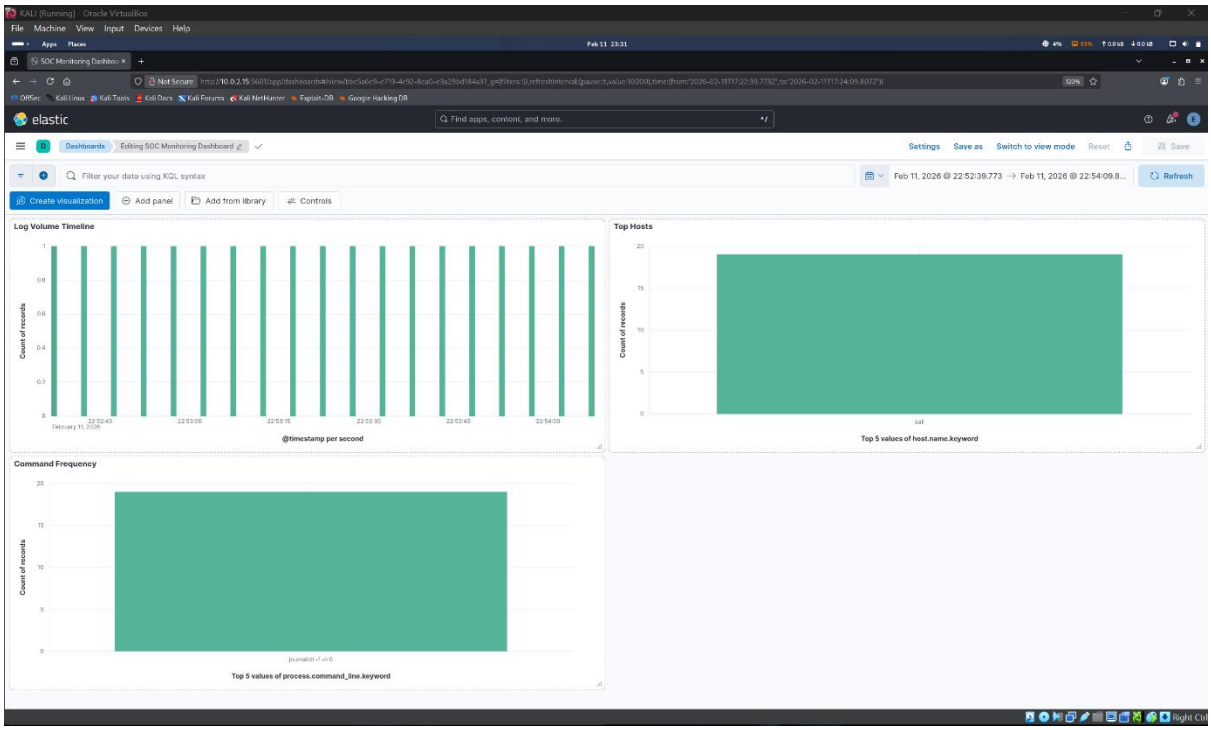
4. Log Ingestion & Data View Creation

Log ingestion was configured using syslog input, and an index pattern was created within Kibana to visualize logs. A timestamp field was selected to allow timeline analysis.



5. Dashboard Visualization

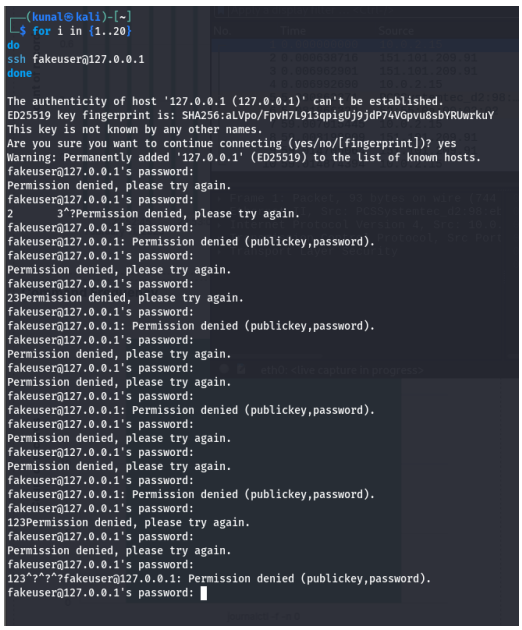
Custom dashboards were created in Kibana to monitor system activity such as log volume and host statistics. These dashboards provided visual insights useful for SOC monitoring.



Simulated Incident & Network Forensics

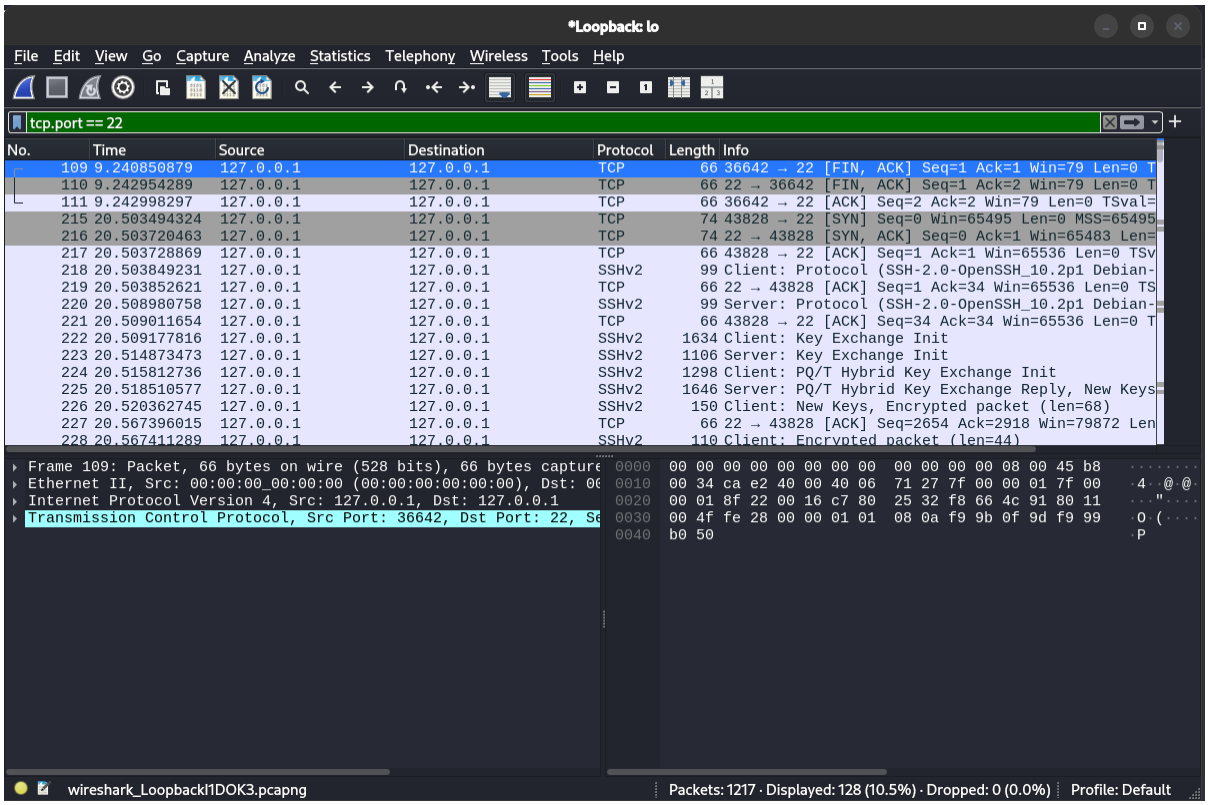
Incident Simulation

A simulated unauthorized SSH login attempt was performed by executing repeated login attempts against the local system. This generated authentication failures to mimic brute-force behaviour.



Packet Capture & Analysis

Network traffic generated during the incident was captured using Wireshark. Packet analysis confirmed communication over TCP protocol targeting SSH port 22.



Attack Vector Identification

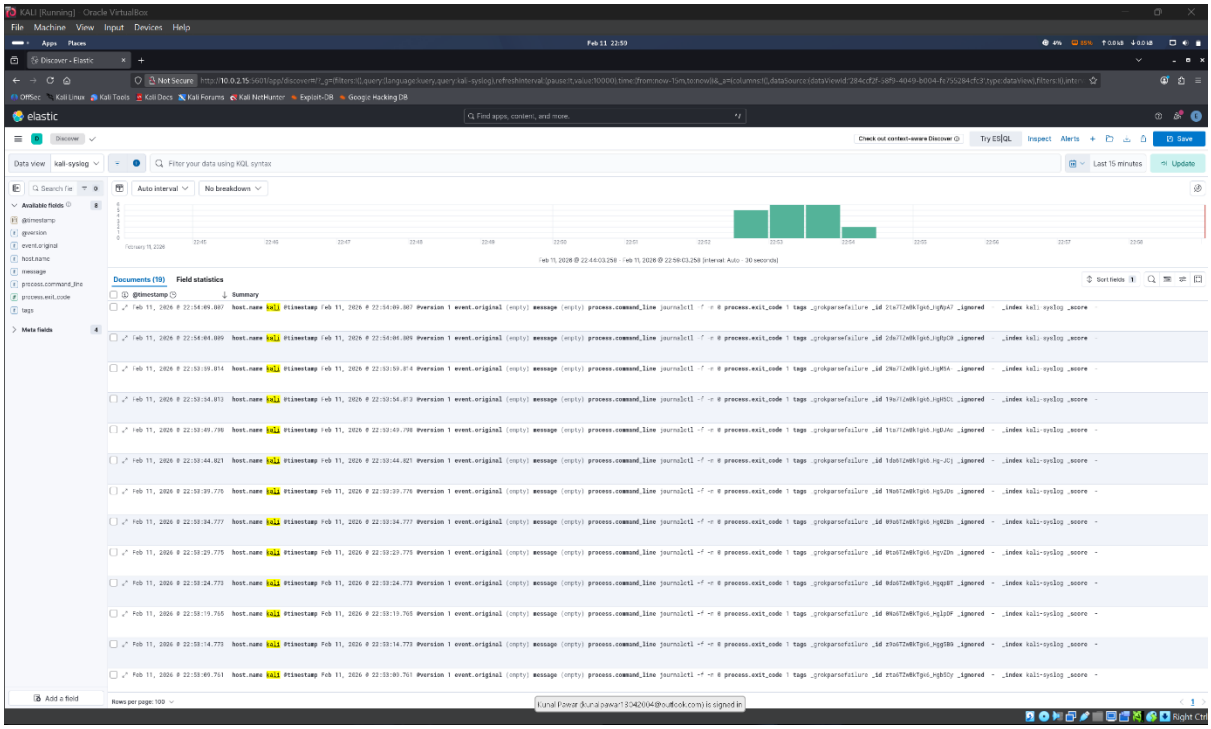
Analysis identified the attack characteristics:

- **Source IP:** 127.0.0.1
- **Target IP:** 127.0.0.1
- **Protocol:** TCP
- **Port:** 22
- **Attack Type:** Repeated authentication attempts

These indicators confirmed a simulated brute-force login attempt.

SIEM Correlation

Logs generated by the incident were visible within Kibana's Discover interface. Timeline correlation showed authentication failures matching captured network activity.



Threat Hunting

Methodology:

Threat hunting was performed by reviewing dashboards and querying SIEM logs to identify suspicious behavior patterns such as login anomalies and command frequency spikes.

Indicators of Compromise (IOC):

The following suspicious indicators were observed:

- Repeated failed login attempts
- Authentication denial messages
- Abnormal command repetition

These patterns indicate potential unauthorized access attempts.

Findings:

Correlation of network captures and SIEM logs demonstrated how centralized monitoring can detect abnormal authentication patterns. The investigation highlighted the effectiveness of combining packet inspection and log analytics for proactive security monitoring.

Conclusion

This task provided hands-on experience across multiple aspects of networking and security operations, including subnet design, traffic monitoring, protocol troubleshooting, SIEM deployment, incident investigation, and threat hunting. Practical implementation helped demonstrate how structured IP planning, packet analysis, and systematic troubleshooting contribute to maintaining reliable network communication. The ELK Stack setup enabled centralized log monitoring and visualization, which supported the investigation of simulated security incidents through correlation of network and system data. Overall, the exercise strengthened understanding of real-world SOC workflows and highlighted the importance of proactive monitoring, forensic analysis, and defensive response in modern cybersecurity environments.